



DIGITAL VLSI DESIGN

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DIGITAL VLSI DESIGN

PART 1

MOS Inverters (Static Characteristics)

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MOS Inverters (Static Characteristics)

Part e - CMOS Inverter

Vinay Reddy Narayana

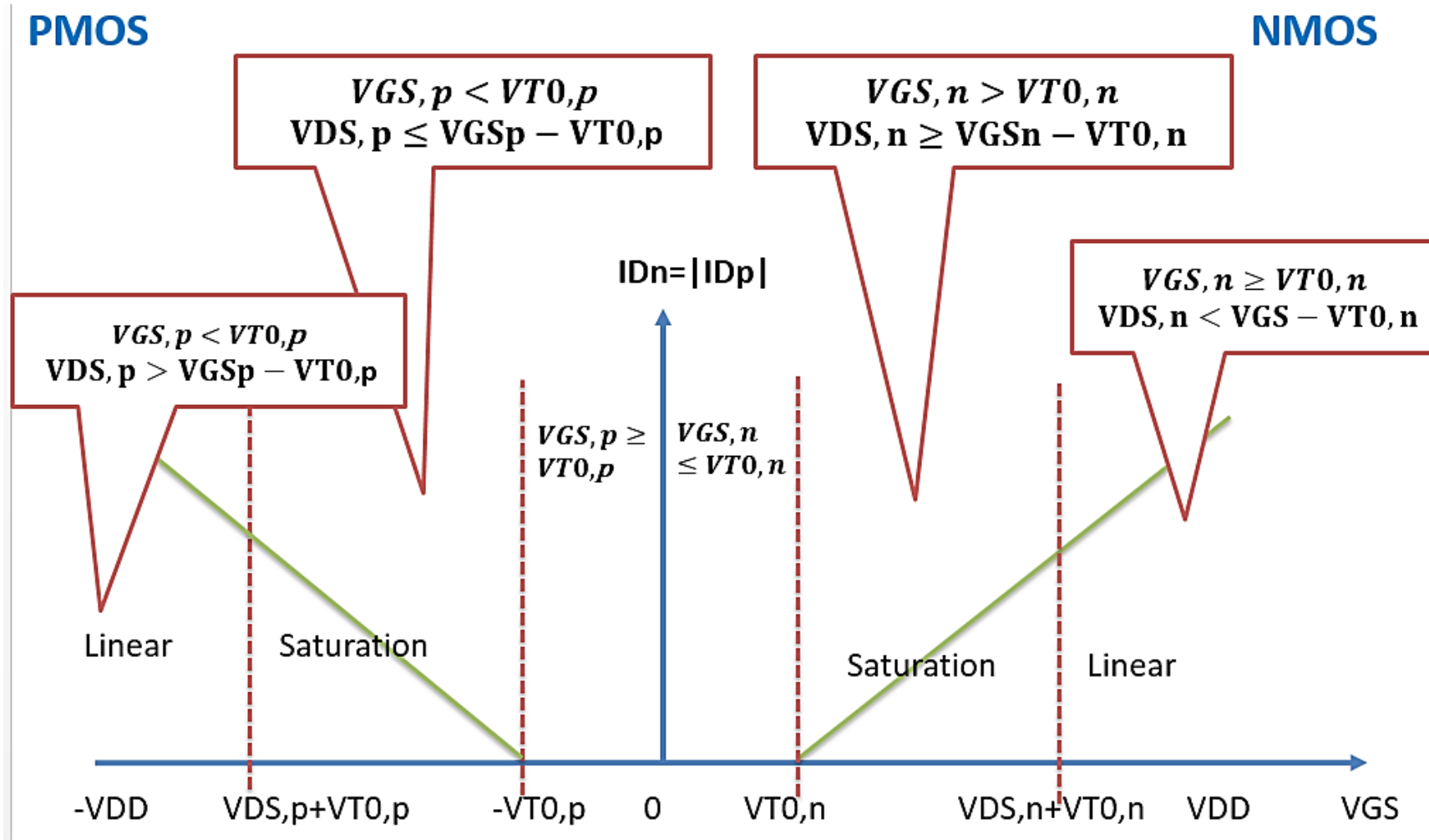
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MOS Inverters (Static Characteristics)

pmos and nmos characteristics

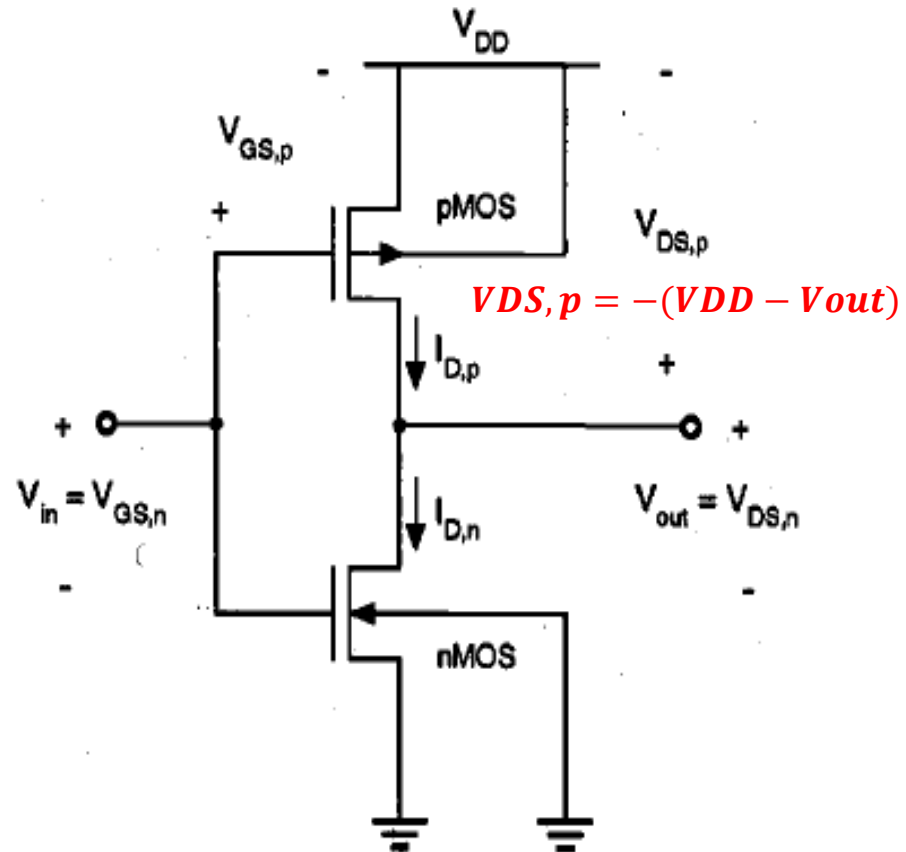
PMOS

NMOS



CMOS Inverter

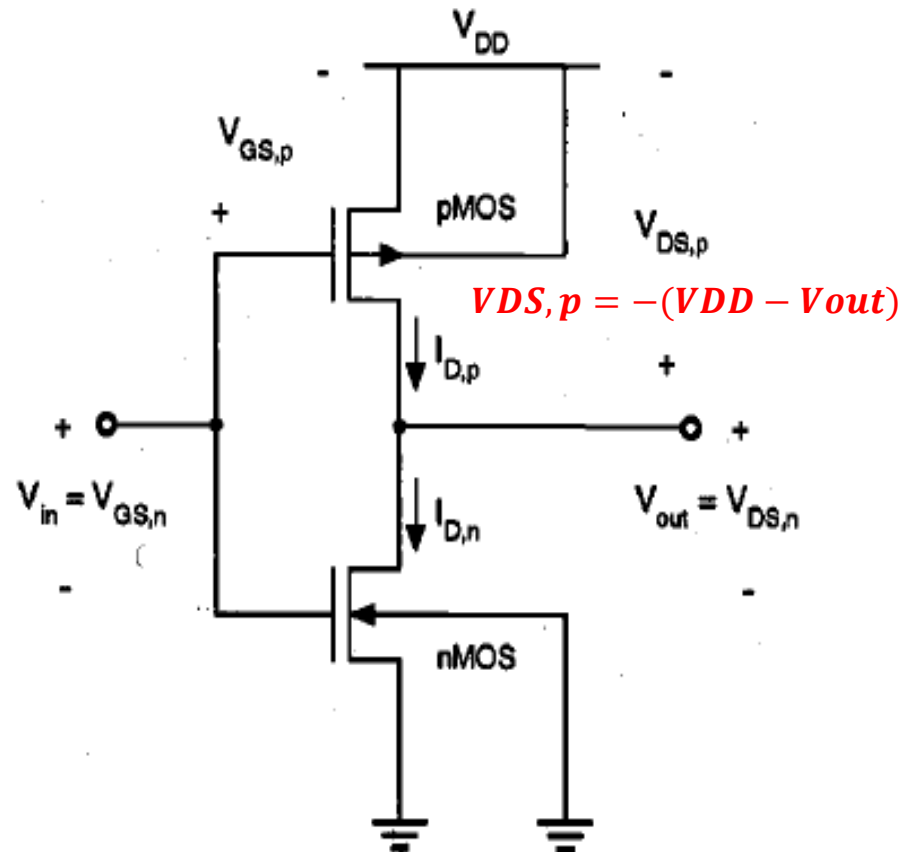
$$V_{GS,p} = -(V_{DD} - V_{in})$$



MOS Inverters (Static Characteristics)

CMOS Inverter

$$V_{GS,p} = -(V_{DD} - V_{in})$$



Condition of Saturation Region of operation

Nmos

$$V_{GS,n} > V_{T0,n} \quad \text{i.e., } V_{in} > V_{T0,n}$$

$$V_{DS,n} \geq V_{GS,n} - V_{T0,n}$$

i.e., $V_{out} \geq V_{in} - V_{T0,n}$

pmos

$$V_{GS,p} < V_{T0,p}$$

$$V_{in} - V_{DD} < V_{T0,p}$$

$$V_{in} < (V_{DD} + V_{T0,p})$$

$$V_{DS,p} \leq V_{GS,n} - V_{T0,n}$$

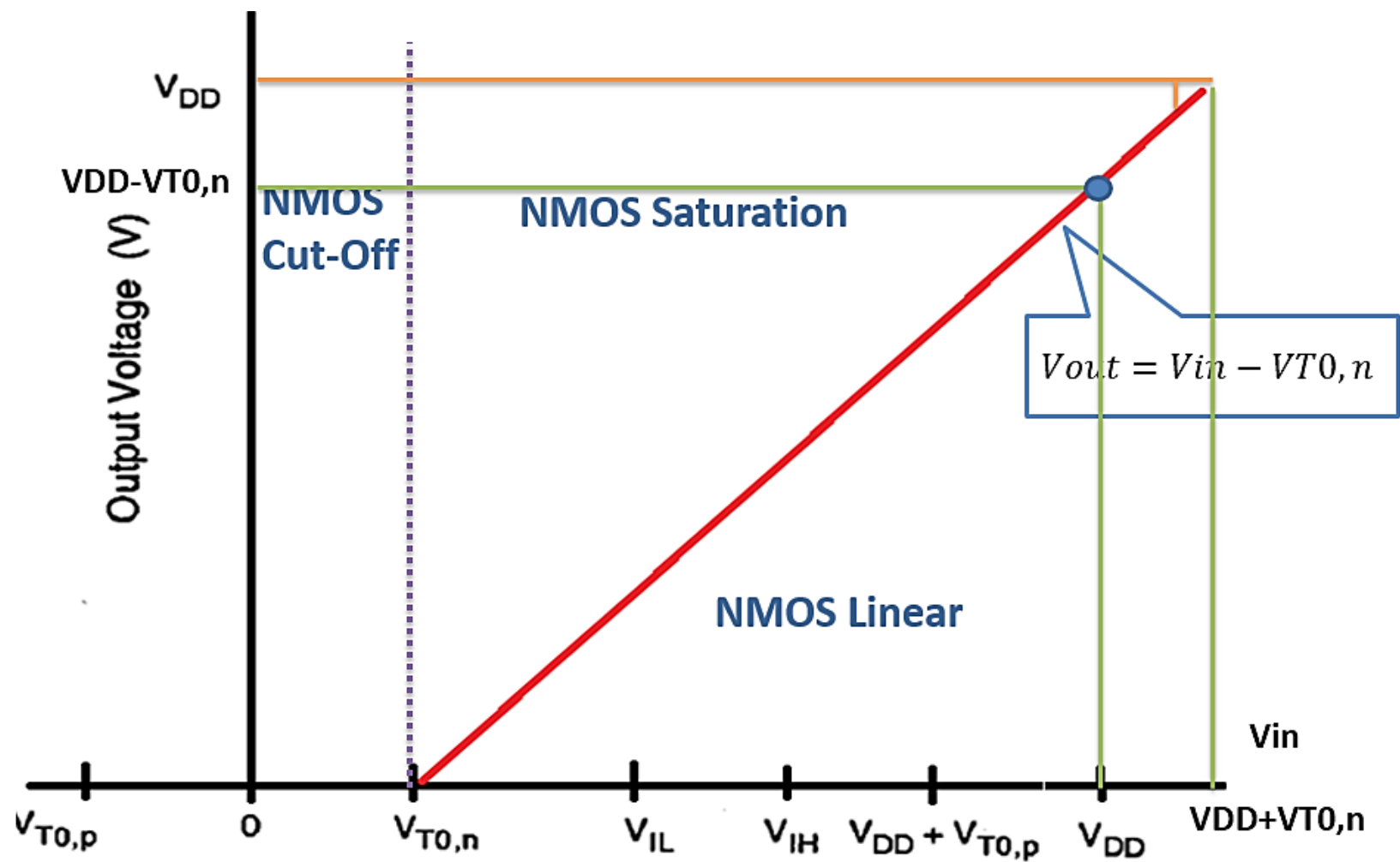
i.e., $V_{out} - V_{DD} \leq -(V_{DD} - V_{in}) - V_{T0,n}$

$$V_{out} \leq (V_{in} - V_{T0,n})$$

MOS Inverters (Static Characteristics)

$$V_{in} > V_{T0,n}$$

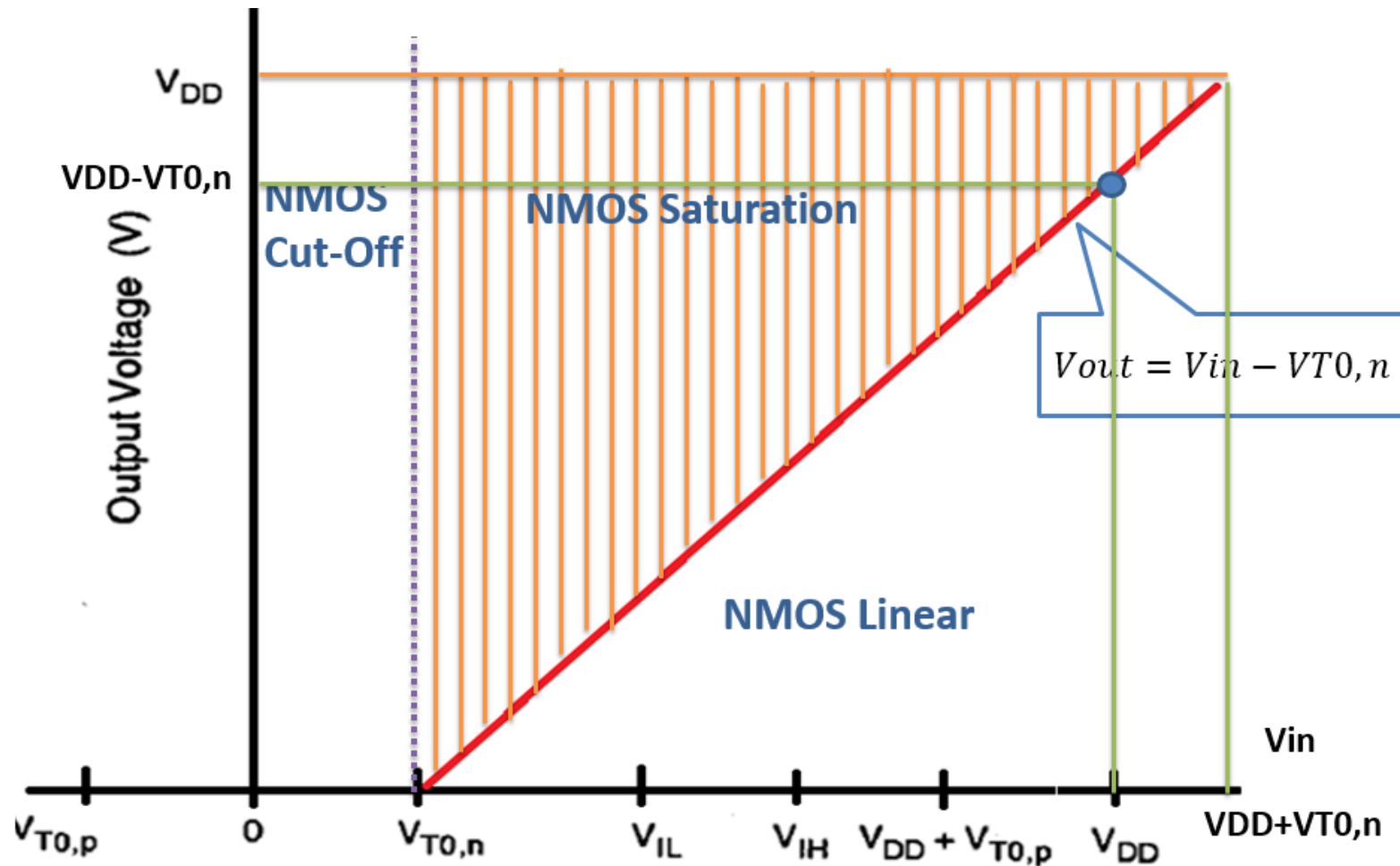
$$V_{out} \geq V_{in} - V_{T0,n}$$



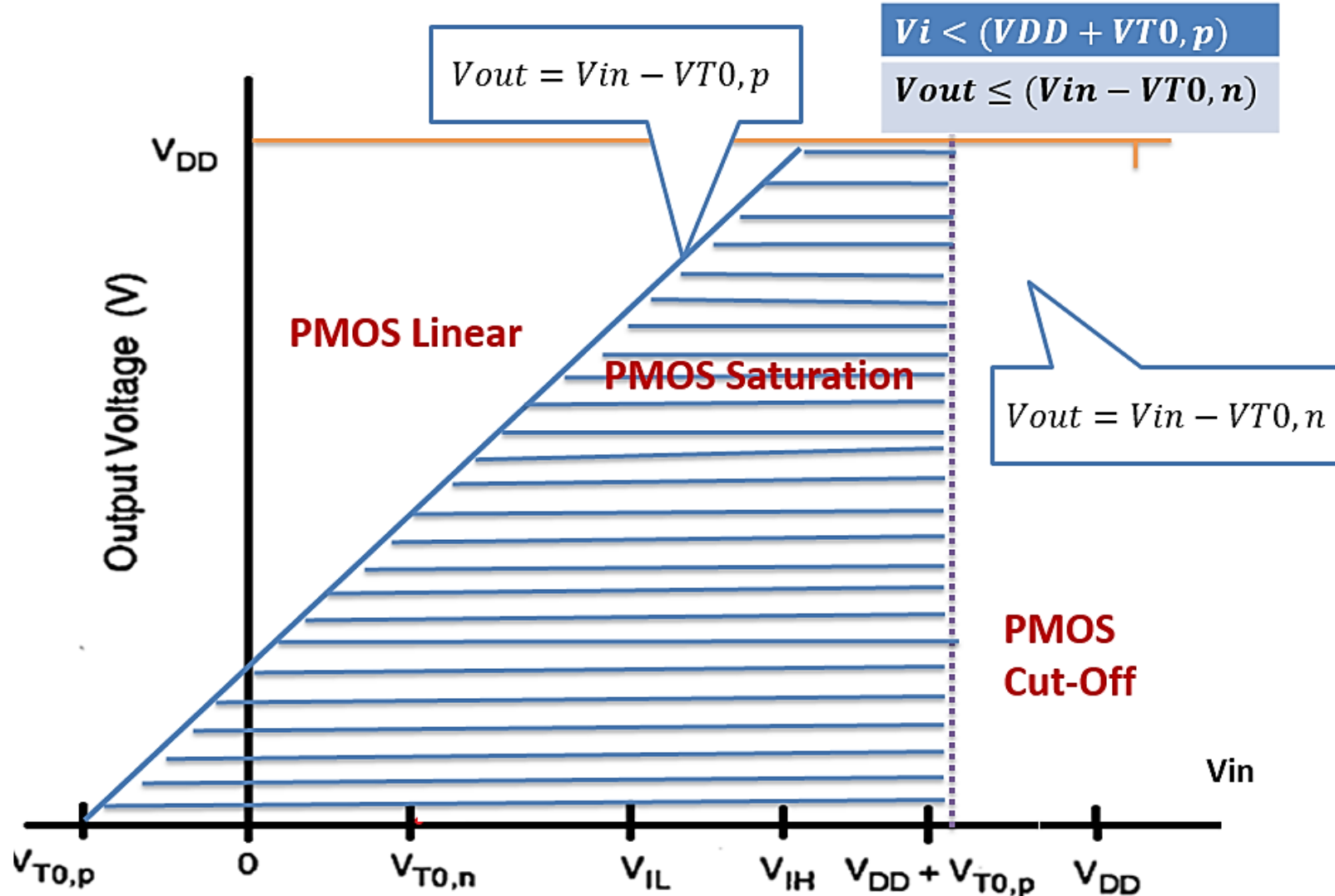
MOS Inverters (Static Characteristics)

$$V_{in} > V_{T0,n}$$

$$V_{out} \geq V_{in} - V_{T0,n}$$

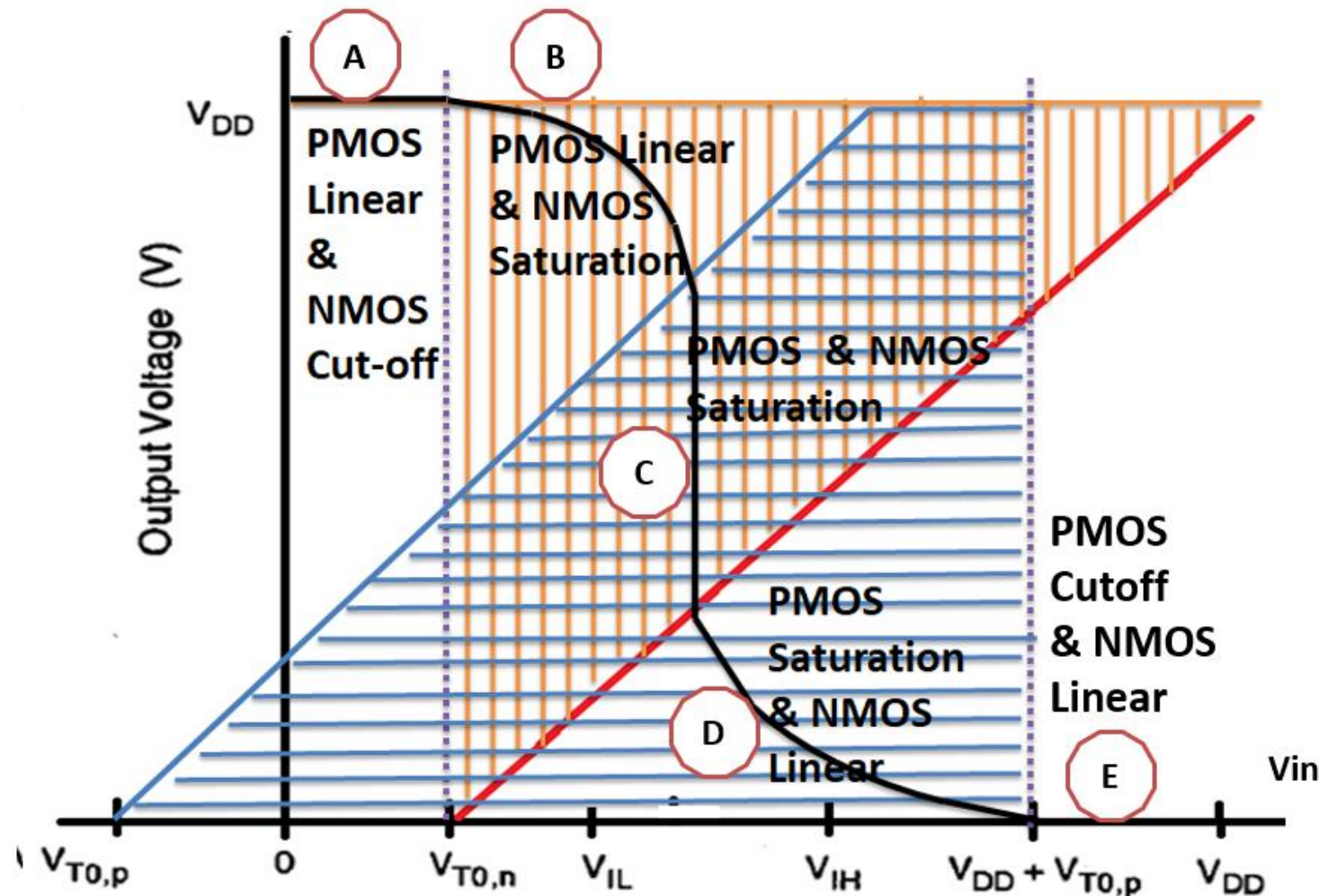


MOS Inverters (Static Characteristics)



MOS Inverters (Static Characteristics)

CMOS Inverter : Region of Operations for nmos and pmos



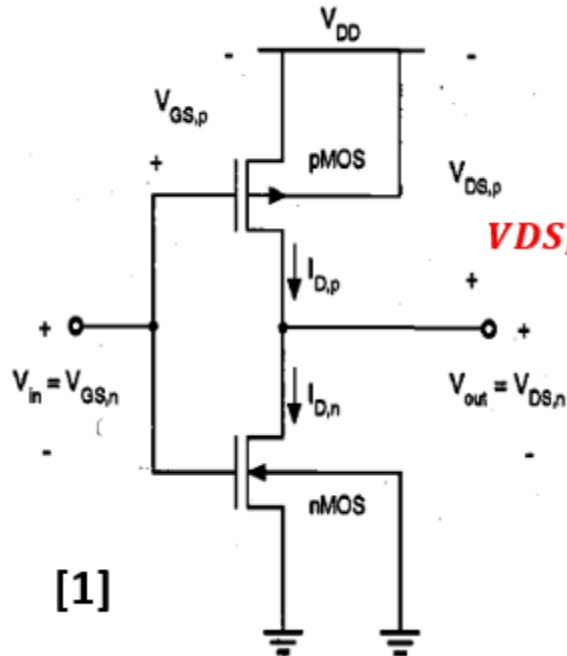
MOS Inverters (Static Characteristics)

Region	V_{in}	V_{out}	nMOS	pMOS
A	$< V_{T0,n}$	V_{OH}	cut-off	linear
B	V_{IL}	high $\approx V_{OH}$	saturation	linear
C	V_{th}	V_{th}	saturation	saturation
D	V_{IH}	low $\approx V_{OL}$	linear	saturation
E	$> (V_{DD} + V_{T0,p})$	V_{OL}	linear	cut-off

MOS Inverters (Static Characteristics)

Calculation of V_{IL}

$$V_{GS,p} = -(V_{DD} - V_{in})$$



$$V_{DS,p} = -(V_{DD} - V_{out})$$

Region	V_{in}	V_{out}	nMOS	pMOS
A	$< V_{T0,n}$	V_{OH}	cut-off	linear
B	V_{IL}	high $\approx V_{OH}$	saturation	linear
C	V_{th}	V_{th}	saturation	saturation
D	V_{IH}	low $\approx V_{OL}$	linear	saturation
E	$> (V_{DD} + V_{T0,p})$	V_{OL}	linear	cut-off

$$\frac{kn}{2} (V_{GS,n} - V_{T0,n})^2 = \frac{kp}{2} [2(V_{GS,p} - V_{T0,p})V_{DS,p} - V_{DS,p}^2] \quad (1)$$

Calculation of V_{IL}

$$\frac{kn}{2} (V_{gs,n} - VT_{0,n})^2 = \frac{kp}{2} [2(V_{gs,p} - VT_{0,p})V_{ds,p} - V_{ds,p}^2] \quad (1)$$

$$V_{gs,n} = V_{in}, \quad V_{gs,p} = V_{in} - V_{DD}, \quad V_{ds,p} = (V_{out} - V_{DD})$$

$$\frac{kn}{2} (V_{in} - VT_{0,n})^2 = \frac{kp}{2} [2(V_{in} - V_{DD} - VT_{0,p})(V_{out} - V_{DD}) - (V_{out} - V_{DD})^2]$$

Differentiate above eqn w.r.t V_{in} as V_{IL} is the voltage defined at slope = (-1)

$$kn \cdot 2(V_{in} - VT_{0,n}) = kp \left[2(V_{in} - V_{DD} - VT_{0,p}) \cdot \frac{dV_{out}}{dV_{in}} + 2(V_{out} - V_{DD}) - 2(V_{out} - V_{DD}) \right]$$

Calculation of V_{IL}

$$\begin{aligned} & kn \cdot 2(V_{in} - VT_{0,n}) \\ &= kp \left[2(V_{in} - VDD - VT_{0,p}) \cdot \frac{dV_{out}}{dV_{in}} + 2(V_{out} - VDD) - 2(V_{out} - VDD) \cdot \frac{dV_{out}}{dV_{in}} \right] \end{aligned}$$

$$\text{Substitute } \frac{dV_{out}}{dV_{in}} = -1$$

$$\begin{aligned} & kn \cdot 2(V_{in} - VT_{0,n}) \\ &= kp [2(V_{in} - VDD - VT_{0,p}) \cdot (-1) + 2(V_{out} - VDD) - 2(V_{out} - VDD) \cdot (-1)] \end{aligned}$$

$$kn \cdot (V_{in} - VT_{0,n}) = kp [-V_{in} + VDD + VT_{0,p} + V_{out} - VDD + V_{out} - VDD]$$

$$kn \cdot (V_{IL} - VT_{0,n}) = kp [2V_{out} - V_{IL} + VT_{0,p} - VDD]$$

$$kn \cdot V_{IL} + kpV_{IL} = kn \cdot VT_{0,n} + kp [2V_{out} + VT_{0,p} - VDD]$$

$$V_{IL} = \left[\frac{kp(2V_{out} + VT_{0,p} - VDD) + knVT_{0,n}}{(kp + kn)} \right]$$

MOS Inverters (Static Characteristics)

Calculation of V_{IL}

$$V_{IL} = \left[\frac{kp(2V_{out} + VT0,p - VDD) + knVT0,n}{(kp + kn)} \right]$$

Dividing Nr and Dr by kp

$$V_{IL} = \left[\frac{(2V_{out} + VT0,p - VDD) + (kR)VT0,n}{(1 + kn/kp)} \right]$$

$$kR = \left(\frac{kn}{kp} \right)$$

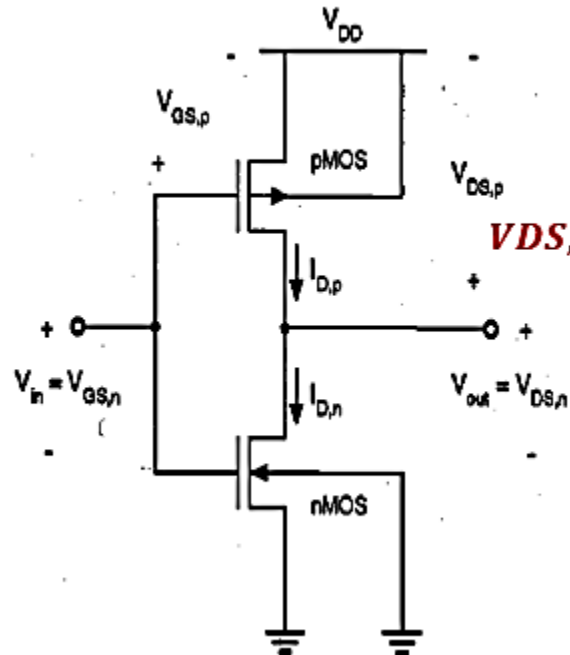
$$V_{IL} = \left[\frac{(2V_{out} + VT0,p - VDD) + (kn/kp)VT0,n}{(1 + kR)} \right] \quad -(3)$$

Must Solve eqn (3) along with eqn (1) to get solution for V_{IL} and V_{out}

MOS Inverters (Static Characteristics)

Calculation of V_{IH}

$$V_{GS,p} = -(V_{DD} - V_{in})$$



$$V_{DS,p} = -(V_{DD} - V_{out})$$

Region	V_{in}	V_{out}	nMOS	pMOS
A	$< V_{T0,n}$	V_{OH}	cut-off	linear
B	V_{IL}	high $\approx V_{OH}$	saturation	linear
C	V_{th}	V_{th}	saturation	saturation
D	V_{IH}	low $\approx V_{OL}$	linear	saturation
E	$> (V_{DD} + V_{T0,p})$	V_{OL}	linear	cut-off

Calculation of V_{IH}

$$\frac{k_n}{2} [2(V_{gs,n} - V_{T_{0,n}})V_{ds,n} - V_{ds,n}^2] = \frac{k_p}{2} (V_{gs,p} - V_{T_{0,p}})^2$$

$$\frac{k_n}{2} [2(V_{in} - V_{T_{0,n}})V_{out} - V_{out}^2] = \frac{k_p}{2} (V_{in} - V_{DD} - V_{T_{0,p}})^2 \quad (1)$$

Differentiate eqn (1) w.r.t V_{in}

$$k_n [2(V_{in} - V_{T_{0,n}}) \left(\frac{dV_{out}}{dV_{in}} \right) + 2V_{out} - 2V_{out} \left(\frac{dV_{out}}{dV_{in}} \right)] = (k_p) \cdot 2 \cdot (V_{in} - V_{DD} - V_{T_{0,p}}) \quad (2)$$

Substitute $\left(\frac{dV_{out}}{dV_{in}} \right) = -1$ and $V_{in} = V_{IH}$, Rearrange

$$k_n [2(V_{IH} - V_{T_{0,n}}) \cdot (-1) + 2V_{out} - 2V_{out} \cdot (-1)] = (k_p) \cdot 2 \cdot (V_{IH} - V_{DD} - V_{T_{0,p}})$$

$$k_n [(V_{IH} - V_{T_{0,n}}) \cdot (-1) + V_{out} - V_{out} \cdot (-1)] = (k_p) \cdot (V_{IH} - V_{DD} - V_{T_{0,p}})$$

$$\left(\frac{k_n}{k_p} \right) [-V_{IH} + V_{T_{0,n}} + 2V_{out}] = (V_{IH} - V_{DD} - V_{T_{0,p}})$$

$$\left[V_{IH} + \left(\frac{k_n}{k_p} \right) V_{IH} \right] = \left(\frac{k_n}{k_p} \right) \cdot (2V_{out} + V_{T_{0,n}}) + V_{DD} + V_{T_{0,p}}$$

MOS Inverters (Static Characteristics)

$$[V_{IH} + kRV_{IH}] = kR \cdot (2V_{out} - V_{T0,n}) + V_{DD} + V_{T0,p}$$

$$V_{IH} = \left[\frac{kR \cdot (2V_{out} + V_{T0,n}) + V_{DD} + V_{T0,p}}{(1 + kR)} \right] \text{-----(3)}$$

Must Solve eqn (3) along with eqn (1) to get solution for V_{IH} and V_{out}

MOS Inverters (Static Characteristics)

Calculation of V_{th}

$$\frac{k_n}{2} (V_{gs, n} - V_{T_{0, n}})^2 = \frac{k_p}{2} (V_{gs, p} - V_{T_{0, p}})^2$$

$$\frac{k_n}{2} (V_{in} - V_{T_{0, n}})^2 = \frac{k_p}{2} (V_{in} - V_{DD} - V_{T_{0, p}})^2 \quad (1)$$

By taking Square root on both side & By definition we get $\mp$ on RHS

$$(V_{in} - V_{T_{0, n}}) = \mp \sqrt{\frac{K_p}{K_n}} (V_{in} - V_{DD} - V_{T_{0, p}})$$

Choose -VE sign in order to get physically correct solution (Note: If + VE is chosen the denominator will become zero which is not physically correct solution for $V_{in}=V_{th}$)

$$(V_{in} - V_{T_{0, n}}) = -\sqrt{\frac{K_p}{K_n}} V_{in} - \sqrt{\frac{K_p}{K_n}} (-V_{DD} - V_{T_{0, p}})$$

$$\left(V_{in} + \sqrt{\frac{K_p}{K_n}} V_{in} \right) = V_{T_{0, n}} + \sqrt{\frac{K_p}{K_n}} (V_{DD} + V_{T_{0, p}})$$

Calculation of V_{th}

$$V_{in} \left(1 + \sqrt{\frac{1}{KR}} \right) = V_{T0,n} + \sqrt{\frac{1}{KR}} (V_{DD} + V_{T0,p})$$

$$\text{Where } K_R = \frac{k_n}{k_p} A = \pi r^2$$

$$V_{in} = V_{th} = \frac{V_{T0,n} + \sqrt{\frac{1}{KR}} (V_{DD} + V_{T0,p})}{\left(1 + \sqrt{\frac{1}{KR}} \right)} \text{ ---- (2)}$$

V_{th} is the threshold voltage of CMOS inverter where V_{in} crosses V_{out} curve. The value of V_{th} defines whether the Inverter designed is symmetric/ Non symmetric Inverter

CMOS Inverter is said Symmetric when $V_{th} = \left(\frac{V_{DD}}{2} \right)$.

In order to have $V_{th} = \left(\frac{V_{DD}}{2} \right)$ K_R should be equal to UNITY and

$$V_{T0,n} = |V_{T0,p}|$$

MOS Inverters (Static Characteristics)

Calculation of V_{th}

By substituting $V_{T0,n} = |V_{T0,p}|$ and $K_R = 1$ in equation 1, We get

$$V_{th} = \frac{V_{T0,n} + \sqrt{\frac{1}{1}} (V_{DD} + V_{T0,p})}{\left(1 + \sqrt{\frac{1}{1}}\right)} = \left(\frac{V_{DD}}{2}\right) \text{ ---- (3)}$$

For symmetric condition $K_R=1$,

$$1 = \left(\frac{k_n}{k_p}\right) = \left(\frac{\mu_n \cdot \epsilon \cdot \phi \cdot \left(\frac{W}{L}\right)_{nmos}}{\mu_p \cdot \epsilon \cdot \phi \cdot \left(\frac{W}{L}\right)_{pmos}}\right) \text{ By considering typical values of } \mu_n \text{ and } \mu_p$$

$$1 = \left(\frac{580 \left(\frac{W}{L}\right)_{nmos}}{230 \left(\frac{W}{L}\right)_{pmos}}\right) \simeq 2.5 \left(\frac{\left(\frac{W}{L}\right)_{nmos}}{\left(\frac{W}{L}\right)_{pmos}}\right)$$

$$\left(\frac{W}{L}\right)_{pmos} \simeq 2.5 \left(\frac{W}{L}\right)_{nmos} \text{ ---- (4)}$$

For a symmetric CMOS inverter with $V_{T0,n} = |V_{T0,p}|$ and $k_R = 1$, the critical voltage V_{IL} can be found,

$$V_{IL} = \frac{1}{8} \cdot (3 V_{DD} + 2 V_{T0,n})$$

Also, the critical voltage V_{IH} is found as

$$V_{IH} = \frac{1}{8} \cdot (5 V_{DD} - 2 V_{T0,n})$$

Note that the sum of V_{IL} and V_{IH} is always equal to V_{DD} in a symmetric inverter.

$$V_{IL} + V_{IH} = V_{DD}$$

The noise margins NM_L and NM_H for this symmetric CMOS inverter are now calculated using (5.3) and (5.4).

The noise margins NML and NMH for this symmetric CMOS inverter are now calculated

$$NM_L = V_{IL} - V_{OL} = V_{IL}$$

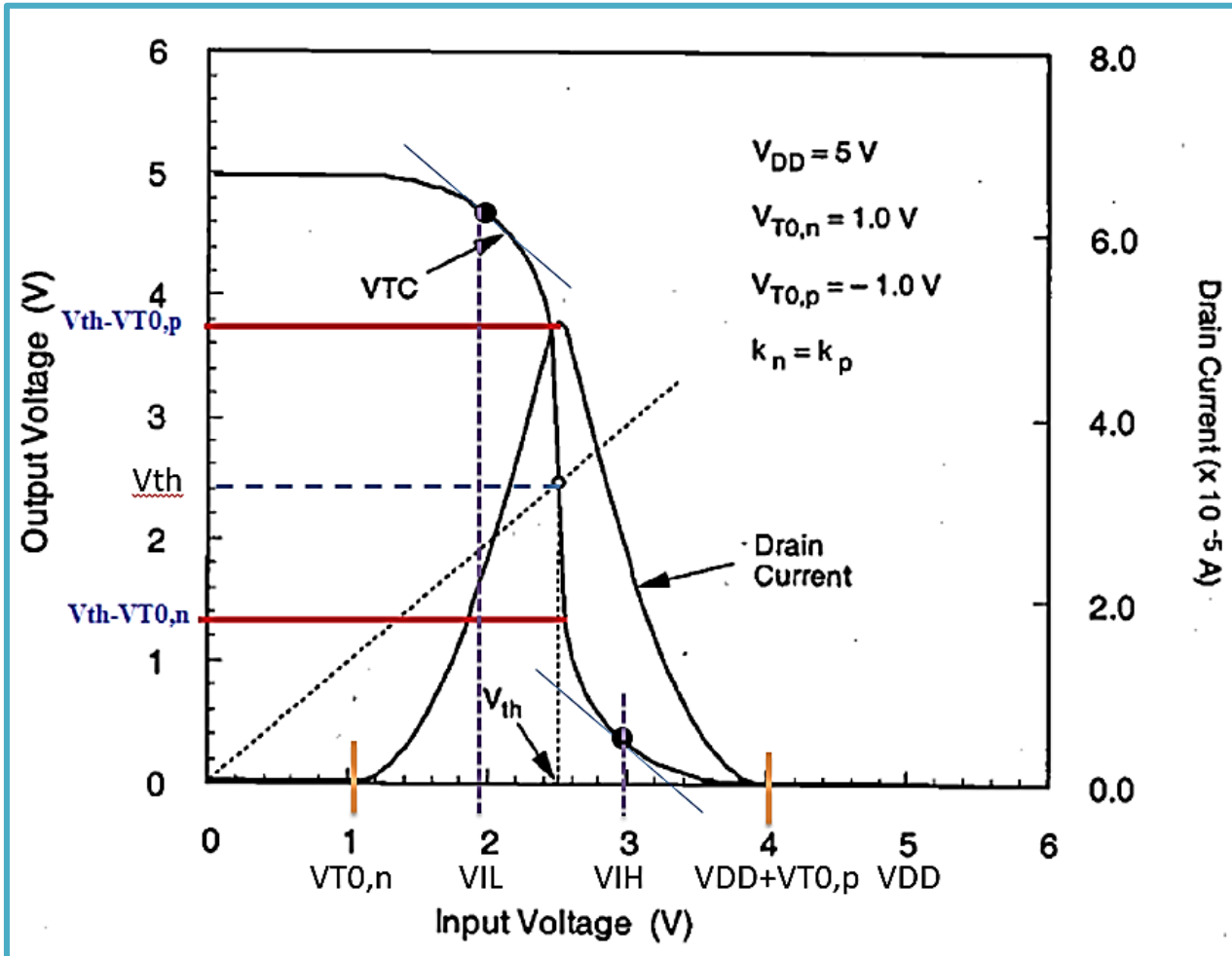
$$NM_H = V_{OH} - V_{IH} = V_{DD} - V_{IH}$$

which are equal to each other, and also to V_{IL} .

$$NM_L = NM_H = V_{IL}$$

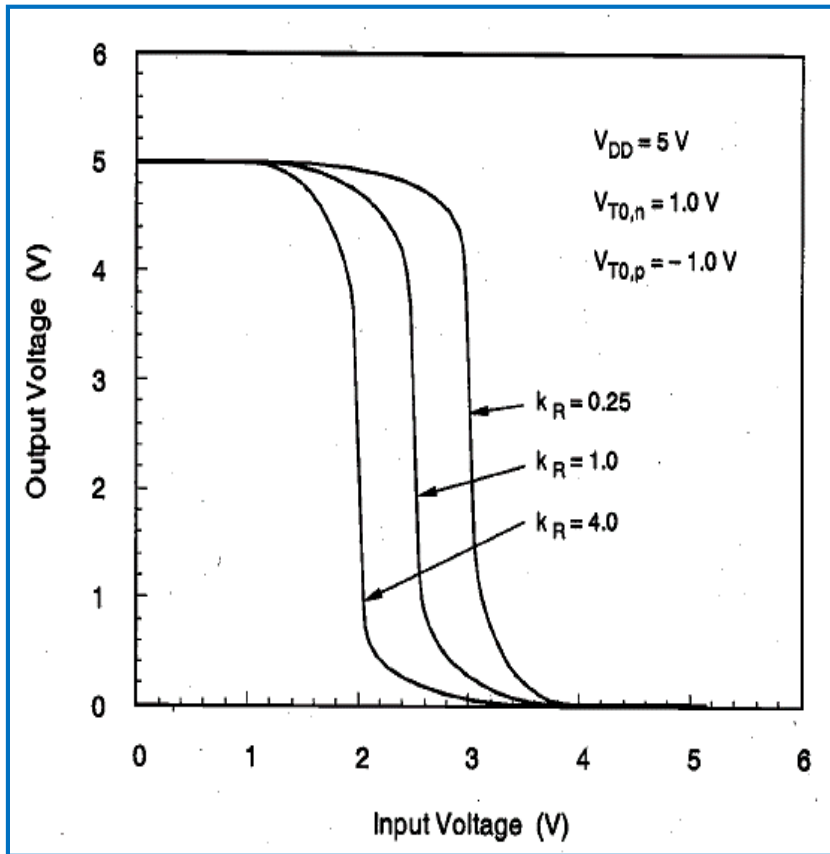
MOS Inverters (Static Characteristics)

VTC-Power Supply current



MOS Inverters (Static Characteristics)

Effect K_R on VTC



$$V_{th} = \frac{V_{T0,n} + \sqrt{\frac{1}{KR}} (V_{DD} + V_{T0,p})}{\left(1 + \sqrt{\frac{1}{KR}}\right)}$$

When $V_{DD} = 3V$, $V_{T0,n} = 0.7$ & $V_{T0,p} = -0.7$

i) $KR = 1$; $V_{th} = \frac{0.7 + \sqrt{\frac{1}{1}}(3 - 0.7)}{\left(1 + \sqrt{\frac{1}{1}}\right)} = \frac{V_{DD}}{2} = 1.5$

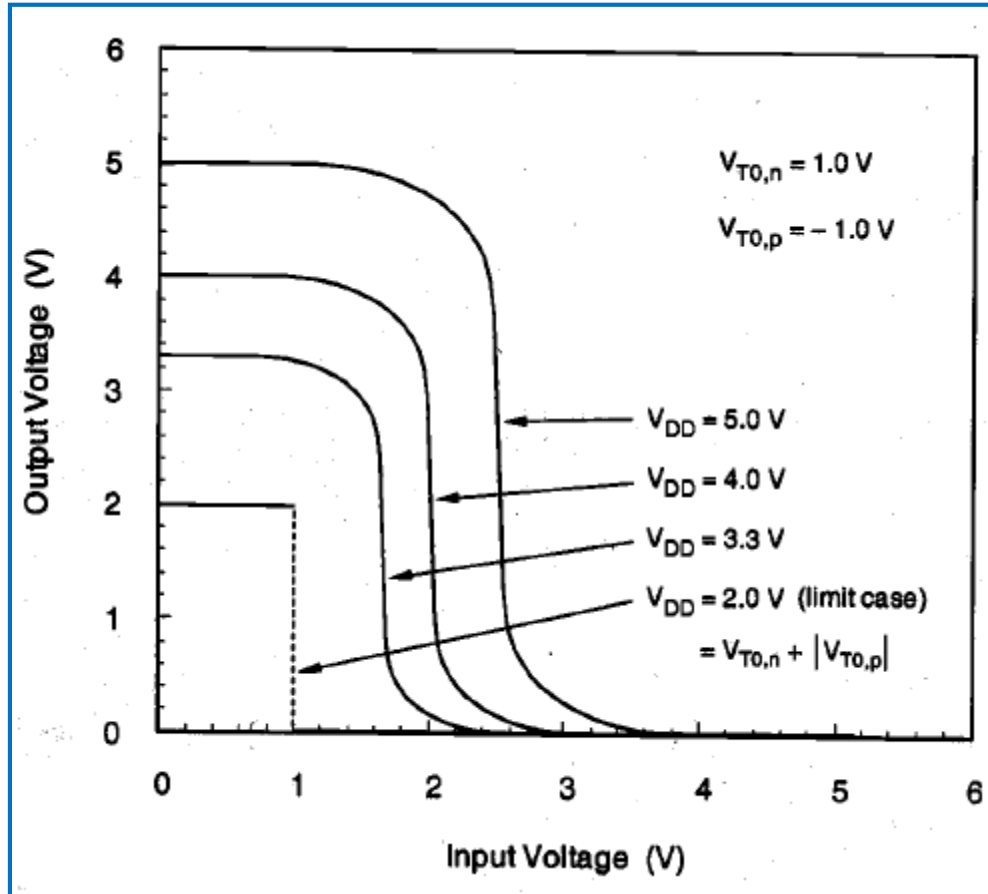
ii) $KR = 4$; $V_{th} = \frac{0.7 + \sqrt{\frac{1}{4}}(3 - 0.7)}{\left(1 + \sqrt{\frac{1}{4}}\right)} = 1.24V$

iii) $KR = 0.25$; $V_{th} = \frac{0.7 + \sqrt{\frac{1}{0.25}}(3 - 0.7)}{\left(1 + \sqrt{\frac{1}{0.25}}\right)} = 1.76V$

From this it clear that INCREASE in KR greater than UNITY will decrease the V_{th} and is always Less than $\frac{V_{DD}}{2}$ & DECREASE in KR less than UNITY will increse the V_{th} and is always greater than $V_{DD}/2$

MOS Inverters (Static Characteristics)

Effect of Supply Voltage Scaling on VTC



*Power dissipation is strong function of V_{DD} .
As a result scaling Power Supply is one of the important criteria in designing.*

CMOS Inverter continues to operate correctly (by neglecting the sub threshold condition) with scaled supply voltage.

The Supply voltage can be scaled down up to minimum value of

$$V_{DD(\min)} = V_{T0,n} + |V_{T0,p}|$$

Correct inverter operation will sustained if at least one of MOS remains conducting for any given input voltage.

MOS Inverters (Static Characteristics)

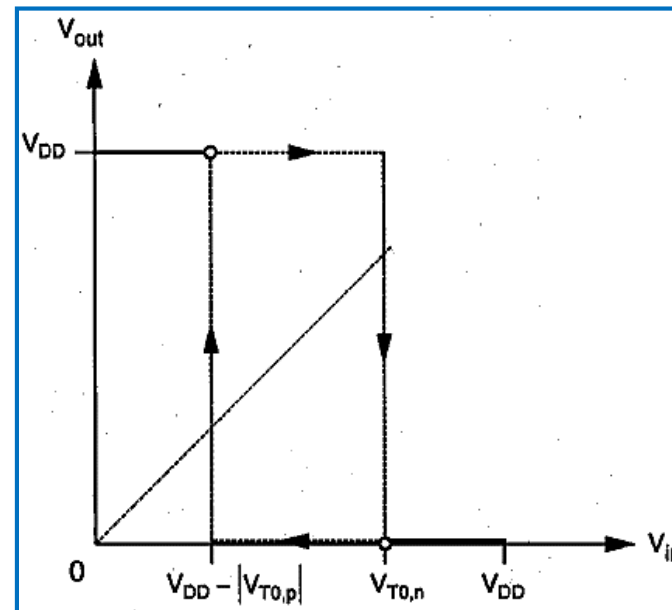
Effect of Supply Voltage Scaling on VTC

If Supply voltage is made
 $V_{DD} < V_{T0,n} + |V_{T0,p}|$

Then, CMOS inverter VTC behaves somewhat differently as it will contain a region in which NONE of the transistors are conducting

The output voltage within this region is then determined by previous state of the output since previous state is always preserved as stored charges at the output node.

The VTC exhibits hysteresis behavior for very Low Supply voltage



MOS Inverters (Static Characteristics)

Example 5.4

Consider a CMOS inverter circuit with the following parameters :

$$V_{DD} = 3.3 \text{ V}$$

$$V_{T0,n} = 0.6 \text{ V}$$

$$V_{T0,p} = -0.7 \text{ V}$$

$$k_n = 200 \mu\text{A/V}^2$$

$$k_p = 80 \mu\text{A/V}^2$$

Calculate the noise margins of the circuit. Notice that the CMOS inverter being considered here has $k_R = 2.5$ and $V_{T0,n} \neq |V_{T0,p}|$; hence, it is not a symmetric inverter.

MOS Inverters (Static Characteristics)

First, the output low voltage V_{OL} and the output high voltage V_{OH} are found, using (5.54) and (5.55), as $V_{OL} = 0$ and $V_{OH} = 5$ V. To calculate V_{IL} in terms of the output voltage, we use (5.62).

$$V_{IL} = \frac{2V_{out} + V_{T0,p} - V_{DD} + k_R V_{T0,n}}{1 + k_R}$$

$$= \frac{2V_{out} - 0.7 - 3.3 + 1.5}{1 + 2.5} = 0.57V_{out} - 0.71$$

Now substitute this expression into the KCL equation (5.59).

$$\frac{k_n}{2} \cdot (V_{in} - V_{T0,n})^2 = \frac{k_p}{2} \cdot \left[2 \cdot (V_{in} - V_{DD} - V_{T0,p}) \cdot (V_{out} - V_{DD}) - (V_{out} - V_{DD})^2 \right] \quad (5.59)$$

MOS Inverters (Static Characteristics)

$$2.5(0.57 V_{out} - 0.71 - 0.6)^2 = 2(0.57 V_{out} - 0.71 - 3.3 + 0.7)(V_{out} - 3.3) - (V_{out} - 3.3)^2$$

This expression yields a second-order polynomial in V_{out} , as follows:

$$0.66 V_{out}^2 + 0.05 V_{out} - 6.65 = 0$$

Only one root of this quadratic equation corresponds to a physically correct solution for V_{out} (i.e., $V_{out} > 0$).

$$V_{out} = 3.14 \text{ V}$$

From this value, we can calculate the critical voltage V_{IL} as:

$$V_{IL} = 0.57 \cdot 3.14 - 0.71 = \underline{\underline{1.08 \text{ V}}}$$

To calculate V_{IH} in terms of the output voltage, use (5.67):

$$\begin{aligned} V_{IH} &= \frac{V_{DD} + V_{T0,p} + k_R \cdot (2 V_{out} + V_{T0,n})}{1 + k_R} \\ &= \frac{3.3 - 0.7 + 2.5(2 V_{out} + 0.6)}{1 + 2.5} = 1.43 V_{out} + 1.17 \end{aligned}$$

Next, substitute this expression into the KCL equation (5.64) to obtain a second-order polynomial in V_{out} .

$$2.5[2(1.43 V_{out} + 1.17 - 0.6) V_{out} - V_{out}^2] = (1.43 V_{out} - 1.43)^2$$

$$2.61 V_{out}^2 + 6.94 V_{out} - 2.04 = 0$$

Again, only one root of this quadratic equation corresponds to the physically correct solution for V_{out} at this operating point, i.e., when $V_{in} = V_{IH}$.

$$V_{out} = 0.27 \text{ V}$$

From this value, we can calculate the critical voltage V_{IH} as:

$$V_{IH} = 1.43 \cdot 0.37 + 1.17 = \underline{\underline{1.55 \text{ V}}}$$

MOS Inverters (Static Characteristics)

Finally, we find the noise margins for low voltage levels and for high voltage levels using (5.3) and (5.4).

$$NM_L = V_{IL} - V_{OL} = 1.08 \text{ V}$$

$$NM_H = V_{OH} - V_{IH} = 1.75 \text{ V}$$

MOS Inverters (Static Characteristics)

5.6 Consider a CMOS inverter with the following parameters:

<i>nMOS</i>	$V_{T0,n} = 0.6 \text{ V}$	$\mu_n C_{ox} = 60 \mu\text{A/V}^2$	$(W/L)_n = 8$
<i>pMOS</i>	$V_{T0,p} = -0.7 \text{ V}$	$\mu_p C_{ox} = 25 \mu\text{A/V}^2$	$(W/L)_p = 12$

Calculate the noise margins and the switching threshold (V_{th}) of this circuit. The power supply voltage is $V_{DD} = 3.3 \text{ V}$.



THANK YOU

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